

ActInf Textbook Group ~ Cohort 2 ~ Meeting 17

(Chapter 7, part 2)

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hello it's February 22nd 2023 and we're in the second discussion of chapter seven in cohort two before we go to the questions of chapter seven and then at the end to look ahead to chapter eight we are going to have Thomas Parr joined for a live stream in five days and so anybody if they would like to raise an area of questions could be conceptual or it could just be a general textbook level or personal question so of course feel free to add anything but just while we're here any thought or like kind of General piece that people are thinking that Thomas could reduce our uncertainty about all right just of course add it when you see fit okay um there were several questions that we've already looked at from seven and several more we can also look a little bit more just again we we looked at the first sections of the chapter itself so does anyone have a chapter 7 question that they want to go to or ADD or do they prefer to walk through the sections in the textbook let's look at equation 7.4 see if we have anything annotated for it okay we don't have the natural language yet so let's just look a little let's pick back up with the um rat in the Maze example go through it connect it to the equations and then return to some to several of the um uh questions down here in seven um so in the earliest part of chapter seven was a passive inference with the example of listening to music now the upstairs action selection part is going to come into play the imperative for Action selection is going to be G like we saw in chapter 5 takes the action prior the prior on different policies which can be understood as like habit and then sharpens it according to just qualitatively provide epistemic and pragmatic value and quantitatively as described in equation 2.6 and 7.4 does anyone want to give a reading or a thought on 7.4 just what is it saying in and of itself and or how does or might this relate to in their words the resolution of the exploration exploitation dilemma yeah I'll go for it foreign we saw before in chapter two I guess um but I think it was in a more comprehensive terms but here we only have a part of that equation which basically deals with the trade-off between a negative um I mean epistemic value and the pragmatic value and how they relate to each other because um as we saw in Chapter 2 the trade-off or the Dilemma

between exploration and exploitation um is basically uh one of the fundamental requirements for any sentient behavior for any sentient agent because as long as the agent tries to uh ex wants to explore the environment in order to informally discover a new um new let's say opportunities or new uh resources be it uh I don't know food resource or any other resource necessary for its survival in the environment it needs to somehow have a trade-off between this kind of having the pragmatic value on one hand namely to survive in the environment and on the other hand to ex to discover new resources for its survival which basically relates to the pragmatic value but either of those two requirements um if taken to uh to its extreme far uh would probably result in the demise of the agent because um so take for instance an agent uh residing in an environment but for a specific a certain period of time it can't survive in that environment using the pre-known resources there but uh when the resources is about to is about um to diminish then that agent need to explore a little bit a little bit further in order to discover some new uh resources and continue it's persistent Through Time word survival yeah great yes Michael oh yeah yeah so so I'm I'm a little bit wondering the the in this epistemic um considerations uh only things can be considered that the agent actually already knows about right and and then I'm wondering why at several places I think it said this in chapter two and here in this chapter also seven two also that um that one would like to find the most unlikely path or one would like to uh to to increase the information value means when yeah um that's what I understood so the past that should be searched for would be the path that maximizes uh information which is uh means it's the the most uncertain path um and correct there's there's aspects that I think are are correct but I just want to draw that out so actually what is being selected for policy are the most likely paths what is an agent like me most likely to do we're sampling our actions from the action posterior so rather than scaling policies by their reward and ranking them by the reward in reward learning and choosing sampling from the reward posterior for example we're going to be sampling from an action posterior so um in the sense that our actions are selected based upon their posterior likelihood we're actually pursuing the most likely path of action indeed the path of least action however we are seeking in the epistemic component especially maximally informative sequences of future observations so in the special case where pragmatic value doesn't matter like we have no preference over outcomes we don't we don't we don't care if our bank account goes up or down then we Implement a pure info Max novelty search in a situation that's totally um known there's no epistemic value to be obtained we have a special case of pure utility however in cases where both utility pragmatic and epistemic are on the table G is proposed to resolve this dilemma by

adjusting the action prior into an action posterior that respects the importance of both of those imperatives that's the unified imperative of action and inference yeah so so so I'm I'm not sure how I should understand this so for instance I want to go to the bakery and buy some bread and then the most informative I mean the more unlikely thing is actually that I go to the barber and cut my hairs so I would con I would because that has more information and going to the Baker and considering what kind of bread I'm going to buy I I I I I go to the barber because that is more that is more informative I'm not sure I understand how I should understand this these values are calculated about generative models not about territories so we again it's it's natural intuitive to want to be like directly mapping from an equation or calculation to a real world territory situation but keep in mind these are um ultimately numerical values calculated for generative model it doesn't make sense to talk about free energy being minimized like for example if a person doing translation simply or of an ant simply foraging it's of the generative model that these values are being calculated on and so if one proposes a generative model of location Behavior so kind of in a previous discussion we had like a higher order like what location am I going to Barber home Baker you know and then we have a lower level model with our footsteps so that's kind of analogous to the live stream 42 slam robotics example then in that situation one could ask how different policy selections were being influenced by their respective epistemic or pragmatic value but it is not the case that we can directly draw from a variable to the territory these are like summary variables or re-weightings on our map and it only makes sense to talk about these summaries on maps not territories Ali actually Michael Levin has an excellent paper from 2019 the computational boundary of a self in which he has an interesting diagram showing the cognitive sphere or cognitive light comes if you will of each cognitive or intelligent agent in which it can have an effect and it's of a more or less a it's concerned so uh it also integrates the yeah that's it so uh so for example um for humans uh because of uh well or highly intelligent uh species or agents uh because of the complexity of its goal-directed activity and agency it results in a kind of larger cognitive boundaries and for uh somewhat uh let's say quote-unquote simpler agents that would result in a much smaller light cone area of cognitive boundaries uh so yeah depending on the complexity of the tasks that the agent is capable of doing that territory of its cognitive effectivity would vary according to the complexity of the thesis these are G is a functional that is going to sharpen our action prior according to such and such if we just wanted to pass through habit we would just pass through habit and um in chapter 5 in the nervous system example this is what's reflected in the bottom here with this dopaminergic circuitry where on the left side of this

basal ganglia or whatever mammalian tissue it happens to be on the left side there's kind of like a habit pass through like e the prior on affordance is basically passed through whereas in G I'm sorry on the right side is sharpening the policy distribution so if we just want to pass through have it we don't even need to apply G however if we want to sharpen our action selection the moment to respect expected free energy in a prospective planning like fashion then we want to calculate $um G$ so what is being calculated with G again this is an equation 2.6 but also here we're taking another look at it so there's two terms looking first at the pragmatic value and the special cases were already mentioned and elsewhere about when one of these or the other is zero okay so fancy e it's an expectation regarding RQ variational distribution that we control of $otilda$ observation sequence conditioned upon policy so we're going to see that both terms are conditioned upon policy so if there's only two affordances then we're only going to compute two different G 's so this is how we kind of prevent like spiraling and Computing like all possible World futures we're conditioning on action and we're interested in we're reducing our uncertainty about which actions to take so that's what this first part is saying so expectation of the variational distribution of upcoming observation sequences conditioned upon actions policies that we can take and it is going to be a natural log to help facilitate just all kinds of things that logs facilitate but it's it's a monotonic relationship with what is being described here and this is about observations conditioned on preferences C so where observations are close to preferences pragmatic value is high where are observations are different than our preferences pragmatic value is low rather than associating pragmatic value with a reward function we're associating it with bounding our surprise natural log self-information surprisal of our preference slash expectation the epistemic value component which can be kind of seen as like a sub function with like a negative I but it's interesting that also that this is kind of a surprisal like um like even pragmatic value enters into an information theoretic surprise like formulation so even our pragmatic value is still informational it's not you know tokens here we see again an expectation over our variational distribution but here rather than on the sequence of observations through time $otilda$ it's the sequence of hidden States Through Time so this is um now looking at underlying latent States conditioned and there's two there's a difference of two entropies then here's P which is like the the kind of actual distribution and then Q the distribution we control now within the parentheses $o tilde$ in both case $o tilde$ conditioned upon States minus $o tilde$ conditioned upon policies so there's definitely a few ways to unpack it and I think we've we've um explored sum but um broadly if the policy is not further reducing your uncertainty about this relationship the epistemic value is

low is reducing your uncertainty about s via o then there's epistemic value I think there's like a lot of open questions that are pretty contextual like how do these implicit or tacit understandings like how does the rat know that going to that Q location is going to reduce its uncertainty how does it know the semantics of the left versus the right cue how does it know what you know this is not the whole story but within the context of doing variational inference on a structurally determined model this is the basis of the separation between pragmatic value defined as alignment between observations and preferences and epistemic value defined as reduction of uncertainty which is to say entropy of how policies sharpen our understanding of the relationship between hidden States and observations Michael yeah so so both of these are expectations so that that is with respect to Future things that might happen both these expectations are actually in the used in a sense of a mean the time Horizon is implicit within π whether that is being calculated whether G is being calculated over π of length one two three Etc in the discrete time formalization but these expectations are actually like center of gravity LaPlace approximation Central limit theorem think expectation of a distribution even though it's like about the future we're actually Computing like the center of gravity or policies of different times but we're not funny enough like like we are but we aren't talking about future time okay so what I maybe don't really understand here is the difference between p and Q I I was of the understanding that P is something that we uh that is present in the in the presence that we experience and P is something that happens in the future now I'm a little bit confused because p and Q both relate here in this equation to the Future right so how can this be understood um the difference between p and Q is not between the present and the future p is like the underlying distribution and Q is our variational approximation so this is like the actual entropy of the relationship between observations and hidden States whereas this is the relationship of the entropy um the entropy of the variational generative models relationship between those two Ali uh yeah I think uh equation 2.3 might tell to somehow clarify that ambiguity between p and Q because in equation 2.3 um yeah that's it the KL Divergence between q and P is defined as the expectation between the surprisal of Q and so the surprise Loop p and one interesting thing about equation 7.4 is that the epistemic value is the expectation with respect to the hidden States but the pragmatic value is the expectation with respect to the observations so we need to keep both of them in mind because here as Daniel just mentioned p is just the uh Q is the variational um I mean the free energy but p is the standard posterior the the standard probability function we apply to a random variable uh so that's why uh they have distinguished between those two um those two functions uh because a

q has uh some um a specific definition so for example in equation 2.2 uh the probability of P without any uh yeah that's it that's just the simple derivation of the probability of uh a random variable being of a specific value is calculated without uh taken into account cues or other parameters servers nice yes the fact that pragmatic value is associated with outcomes is what ties closely to perceptual control theory and the fact that epistemic value is associated with hidden States is related to the avoidal of overfitting we're not reducing our epistemic value on observations observations are being evaluated for their pragmatic value hidden State inferences are being evaluated and and through them policies are being evaluated for their epistemic value yeah so I'm a little bit confused about about this you said that cure relates to the variational free energy but the variation of free energy I mean we made this distinction between the G which is the expected free energy and the F which is the variation of free energy and now inside that expected free energy is a variational free energy and so yeah just to be clear cues the variational distribution this is the guy we control accuse the guy we control P we don't mind your p's and q's Q is the one we control is a real-time functional that's a function of our variational density q and incoming data it doesn't engage with planning expected free energy a host of cousins like free energy of the expected future that Baron millage and others have proposed so this is not like the only way to even do this energy-based method in the future this is rather than a function of um taking in only q and Y as arguments here the primary argument is the policy prior vector and then the computation of this functional involves Q the variational distribution but like Q could be a gaussian so we'd be looking to parameterize the variational parameters you know the median and the variance the two parameters of a gaussian and the true generative process may not be gaussian but we would be looking just like in a linear regression we'd be looking to minimize the sum of squares with the L2 Norm in this variational inference setting we'd be looking to fit the best gaussian mean and variance given the the family of the variational distributions that we had chosen yeah so one other angle can be e is actually the actual posterior probability uh that uh is calculated uh according to the actual States um we have at our disposal but Q is the approximate posterior so it's not the exact posterior that has been calculated in P but yeah Q is related to variation of free energy in the sense that it's deployed in calculating the variation of free energy for the in order for the free energy to be uh attractable parameter uh to be I mean implementable in a computational task we need to use this approximate variation of free energy because otherwise we don't have uh either all the information we need to calculate the exact posteriors or it's too intractable to compute so that's the reason behind using approximate

variational free energy great and I think one kind of as a valiant this idea of like probably approximately correct which is kind of like when they say almost surely in math but more on the statistics side if it's probably approximately correct it's like actually within the context of approximate Bays that is what is going to be not just good enough but like as good as you can kind of aim for matrices are defined a has two slices A1 and A2 are we going to give it as two modalities of a and the dimensionality is clear in the Matlab implementations it's clear in the pi mdp it's not exactly exactly the same syntax but like the structure of these matrices in principle are the same is describing the probabilistic mapping from location to exteroceptive cues so it's not exactly an identity Matrix but basically this is the accurate GPS sensor when it's at this location start location it knows exactly where it is so again you can be like well how does it get this implicit knowledge of where it is right that's part of the complexity of cognitive modeling but suffice to say that we're in a setting where the location is known without error but one can imagine that there would be different outcomes and the need to do different parameterizations if like there were off diagonal elements A2 is the um probability so they sum across to one um or I'm sorry they they call them sum to one the the food location the the between the spatial location and then the outcome of food so semantically this represents beliefs about food location in the a here 7.5 is the same as 7.4 it's just that here it's like two in the upper left 98 on the off here we see 98 on the upper left two on the off here we have one in the third row empty on the second here those two rows have been flipped to represent the block on the left side versus the block on the right side the food can be either on the left or the right side the rat can either make the move to go to the left or the right side or can go to the Q if it decides to go for the Q um not the Q letter but the c-u-e if it decides to receive an informational cue then it will find out where the food is again check out model stream 7.2 with the epistemic foraging and all of that sort of um stuff okay these slices of B are describing movement affordances because again with respect to the hidden State being location B describes how different policy selection Bears upon hidden States changing Through Time C is describing the relative preference for different outcomes the C1 vector is about the first modality five of them that's the locations and there's a negative for the starting location so that's like a little bit of like a like a kind of get going because if it preferred to start location it might just stay there then in the second modality that's referring to this shorter Vector of these one two three and that's these one two three in various modeling settings we've already started to see that like the C scaling matters a lot and so that's one reason why I think it's tentative or hypothetical to say that simply like equation 7.4 resolves explore exploit because it sets up the

space for explore exploit to be framed absolutely it does however depending on how the variables are parameterized you can end up with an agent who still goes for the pragmatic value every time or who still goes for the information value every time so in terms of outcomes it's not simply resolved by knowing what equation 2.5 is but rather this opens up a space of modeling in which adaptive Solutions can be reached so if this was six thousand zero six thousand zero it's like reward is so good it's going to be pursued if this was flat zero zero zero then it would be pure epistemic information so that's why it's so important that we develop statistical power methods and parameter sweeps across simulations because just making one teammates and saying like Oh look The Rat does this or that it's kind of like d also is described for the two modalities and um it um describes the priors on the two there's a belief about the first I'm actually just just I'm a little bit curious why there's five elements here but only four elements here and then in the second modality there which again is referring maybe this is like a a minus there's one fewer um element here because there's only two um here's like reflecting an equal prior over where the food is so a priori is 50 50 left or right so do you go do a 50 50 bet left or right on the food or do you go to get the epistemic cue which of course reduces your uncertainty about where the food is now we're going to dive a little bit deeper into epistemic value itself so in 7.4 they're saying we're going to go ahead and call just the epistemic value decomposition term I equation 7.8 so here is just a separation of our as it was written in the prior equation that is going to be Rewritten in terms of a KL Divergence $q(s)$ and $Pi(q)$ and s and Pi they're the same on both sides here's POS here's $q(o|pi)$ and it's it's definitely worthwhile I think for us to write out what these verbally mean because they have very I think interesting um implications for again like what is the epistemic value and then here is um with this semicolon being noted this is reflecting a pure Information Gain salience Bayesian surprise imperative here's some simulations showing at like time two it goes to the to the get the information and then it proceeds to the left Ali uh yeah just about your previous remark about uh why C has the five elements and D the C is four elements because uh C actually uh corresponds with the the elements of a matrices so because a matrices uh actually describe uh the uh the hidden the states that the environment um uh is dealing with but because D is corresponds with B Matrix and uh because B Matrix is a transition Matrix uh that's why uh we don't need to have a redundant additional element for that so D being priors yet Maps unto B Matrix thanks and right now like you have to like specify the tensor dimensionality and add some labels but they're developing a lot of functions in pi mdp and people who work with it can definitely have a lot of feedback on slightly more intuitive and scalable ways to specify

dimensionality which is currently one of the rate limiting steps in pi mdp application just getting through seven and then just so we can look ahead to eight continuous time so Precision at negative entropy neg entropy this is the high temperature and low temperature high temperature differences are erased low precision low temperature absolute zero High precision differences are accentuated that has technical usage in certain statistical distributions isocate paradigm par and Frist in 2017. isicating is while our eyes may dwell on what we may associate with pragmatically valuable broadly speaking isocating is driven by Information Gain so it's a great place to study how pragmatic semantics at a higher level are in a predictive processing type relationship with isicating at a lower mechanical level with this level almost surely being driven by Information Gain learning a novelty here um one kind of question that was arising was like why is our dashed line between only here but apart from that priors are being provided On A and B so that's sort of like the con again it's like a little bit like wise B over here so it's about the topology not not the geometry in this situation what's happening is that a prior is being provided on a so that these A and B in principle can be learnable how that plays out and how that influences the pseudocode of the action perception Loop Pi mdp is very clear about that um conjugate priors this is facilitating like amongst other um abilities the the dirichlet conjugate prior for the categorical distribution so if we have probability distribution over categorical outcomes is it hot or cold then the dear shlay can be used in an urn counting way like every time you observe a warm day or a cold day you just put in a ball of that color into the urn it could be 2 and 2 and then the next ball is very influential it could be a thousand and a thousand and then the next ball is not so influential and that turns out to be what's called a conjugate prior that enables the categorical distribution to stay as a classical probability axioms and all of that while also allowing the conjugate prior to have this like counting or learning by Counting strategy um inferential approach to learning this is a difference with many other areas they're proposing learning schemes that may or may not be biologically relevant whereas a lot of work has gone into understanding how the perceptual or fast inference and learning or slower parameter learning processes are unified but of distinct time scales and active inference here in 7 11 equation they are going to delve more into like learning as active yes it's about policy everything's being conditioned upon policy there's a lot that one can say about the integration of action so this maze learning task is fundamentally intertwined with the action selection the agents updating of their likelihood is related to their action selection amidst uncertainty they're not seen as like two separated phases here's where they just drop some kind of threads and further literature maybe further versions of the

textbook are going to come into more detail hierarchical deep inference nested modeling um something that people talk and think about a lot but here we see the figure 4.3 or figure 7.3 the rake and the policy selection fractal it's a rake of rakes structure learning and Bayesian model reduction just some model comparison and Analysis techniques that support Bayesian statistics another example involving a hierarchical inference model is described and that has been used in the context of reading and other visual inference tasks that's chapter seven let's quickly look chapter seven was discrete time generative models chapter 8 is going to be continuous time generative models the main setting in which discrete time models have been studied is Central or cognitive decision making categorical decision making in contrast the major setting in which continuous time active inference models to this time have been used in is in continuous movement or motor control tasks so like in live stream 46 active models do not contradict folk psychology with Alex Kiefer Ryan Smith and Maxwell ramstead they talk about like motor AI Mai and decision active inference Dai there's going to be a um Proto Bayesian physics like representation that's going to enable us to do continuous things specifically the separation of a data observation sequence in terms of an underlying function and a fluctuation stochastic term um a latent States derivative with a little Dot change in latent States as a function of the current status of the Hidden State and slowly varying causes which are going to be playing the role of policies in continuous time formulization and again partitioning out of our stochastic fluctuations so actually this part is allowed is this part is guaranteed to have zero mean and that's what facilitates like Bayesian mechanics a lot of other statistics again just like quickly looking through it this is $y = g(x)$ and F relationships continue to be built up to again talk and this is very SPM like here's our actual sensor readers readings from the EEG fmri Meg their function of neural activity neural activity is a hidden latent state it is a function of current neural activity and attention for example that physics analogy in the continuous State space can be extended into Newtonian Dynamics and as Dalton and Bayesian mechanics have laid out in the last year or two it goes way way further than this but this is a spring so this is hooke's law as an as a mere active inference entity this is like a boring active entity that just dampens but it's just to show that we can understand the generalized coordinates of motion which is to say polish position velocity acceleration snap crackle pop Etc et cetera et cetera generalized coordinates of motion within this continuous time framework dynamical systems often which are conditioned to be smooth and differentiable are amenable to continuous time formalizations in the discrete time we'd say well we have you know one two and three and then we're going to be moving to one of the locations in the grid

world and we have a transition Matrix over grid world in the continuous time setting here this is the spinal cord of the butterfly here's the sensory data coming in why and then a descending hidden State and uh error deviation that's being resolved through policy selection in the continuous space so this is hearkening back to chapter 5 and the spinal cord reflex how continuous time generative models can be applied to continuous spinal motor reflexes as opposed to the usually more centrally Associated discrete categorical decision-making Dai there's a sidebar on Precision attention and attenuation there's the discussion of lock Volterra which is also called Predator prey or winner list Dynamics like you can't win in this ecology there's also there are ecologies that collapse of course but activity Dynamics amongst brain regions have been modeled for decades as these winnerless competitions also called neural Darwinism and that's relating to friston's lineage with Edelman more examples of how these kinds of winnerless competitions can be used in the context of continuous time generative models to generate smooth action sequences like this sort of gibberish cursive writing learning is approached in a continuous model everything in chapter 8 is kind of being contrasted or juxtaposed with what we saw at chapter 7 in the discrete time they introduced the Lorenz attractor which is unpacked in significantly more detail in live stream 32 on Markov blankets and stochastic chaos it is basically this example with the Lorenz attractor which is showing that generalized synchrony which isn't lockstep but rather Mutual information amongst couple can happen even when those coupled dynamical systems are Lorenz attractors they don't have to be like both like simple linear systems um so that's like good to know and they've used that to simulate Bird song where before learning they're kind of like on and off a synchronization manifold and then after learning they've sharpened their their um joint distribution again this begs further questions how do they know the semantics of a song and how do they learn the song but this is just an example at this level of complexity um 8.5 Hybrid models here we have the rake within a rake but notice that the outer rake there should be a line connecting this G to Π the outer rake is Dai here we see hidden States Through Time T minus one now future Tale of Two densities observations Through Time classic Dai chapter 7 up top but the nested model has the same structure figure 4.3 but here we see the position velocity acceleration generalized coordinates pattern occurring so this is called a hybrid model because the discrete time model is nesting a continuous time model so this has been used to model things like where do I want to look from a categorical perspective and then here's the oculomotor cicades at a continuous model that's nested within that more equations isucade model summary and

key advances in continuous time models end of the chapter all right thank you we'll take a break and then begin with the next cohort in just two minutes